

Грамматика языка Пфру

Алфавит

```
<letter> := "A"... "Z" | "a"... "z" | "_"  
<rletter> := "A"... "Я" | "a"... "я" | "_"  
<digit> := "0"... "9"  
<any> := ?любойсимвол?  
<identifier> := <letter>{<letter> | <digit>}
```

Литералы

```
<integer_literal> := <digit>{<digit>}  
<float_literal> := <digit>{<digit>} "." <digit>{<digit>}  
<char_literal> := "'" [<letter>] "'"  
<string_literal> := "\"" {<letter>} "\""   
<bool_literal> := "true" | "false"
```

Пробелы и переносы строк

```
<space> := " " | \t  
<newline> := \n | \r | \r\n
```

Типы

```
<primitive_type> := "i8" | "i16" | "i32" | "i64" |  
                    "f32" | "f64" | "char" | "stringa" | "bool"  
<array_type> := <primitive_type> "[" <integer_literal> "]"  
<type> := <primitive_type> | <array_type>
```

Блоки

```
<block> := "{" {<statement>} "}"
```

Операторы

```
<statement> :=  
    <var_decl> ";"  
  | <assignment> ";"  
  | <expr> ";"  
  | <if_stmt>  
  | <while_stmt>  
  | <do_while_stmt>  
  | <for_stmt>  
  | <return_stmt> ";"
```

```
<var_decl> := <identifier> [":" <type>] ["=" <expr>]
```

Ветвление и циклы

```
<if_stmt> := "if" <expr> <block>  
          {"elif" <expr> <block>}
```

```
<while_stmt> := "while" <expr> <block>  
<do_while_stmt> := "do" <block> "while" <expr> ";"
```

```
<for_stmt> := "for" <identifier> "in" <range> <block>
```

```
<range> := "[" <expr> ";" [<expr>] ";" <expr> "]"
```

```
<return_stmt> := "return" <expr>
```

Выражения

```
<expr> := <assignment>  
<assignment> := <identifier> "=" <comma_expr>  
<comma_expr> := <logic_or>{"," <logic_or>}  
<logic_or> := <logic_and>{"||" <logic_and>}  
<logic_and> := <bit_or>{"&&" <bit_or>}  
<bit_or> := <bit_xor>{"|" <bit_xor>}  
<bit_xor> := <bit_and>{"^" <bit_and>}  
<bit_and> := <equality>{"&" <equality>}  
<equality> := <rel>{"==" | "!="} <rel>  
<rel> := <shift>{"<" | ">" | "<=" | ">="} <shift>  
<shift> := <add>{"<<" | ">>"} <add>
```

```

<add> := <mul>{("+" | "-") <mul>}
<mul> := <unary>{("*" | "/" | "%") <unary>}
<unary> := [("+" | "-" | "!")] <primary>

```

Первичные выражения

```

<primary> :=
    <literal>
  | <identifier>
  | <call_expr>
  | <array_literal>
  | "(" <expr> ")"

<call_expr> := <identifier> "(" [<arg_list>] ")"
<arg_list> := <expr>{"," <expr>}

<array_literal> := "{" <expr>{"," <expr>} "}"

<literal> := <integer_literal> | <float_literal> |
             <string_literal> | <char_literal> | <bool_literal>

```

Программа

```

<program> := {<toplevel_decl>}
<toplevel_decl> := <repr_func> | <arrow_block>

```

Функции repr(...)

```

<repr_func> := "repr" <identifier> "(" [<param_list>] ")"
             ["->" <return_type_list>] <block>

<param_list> := <param>{"," <param>}
<param> := <identifier> ":" <type>

<return_type_list> := <type>{"," <type>}

```

Блок стрелок #strelki

```
<arrow_block> := "#" [<identifier>] "{" <arrow_line> "}"  
<arrow_line> := <arrow_node> <arrow_op> <arrow_node> ";"  
<arrow_node> := "start" | "end" | <identifier>  
<arrow_op> := "->" | "-(" <literal_list> ")">  
<literal_list> := <literal> ", " <literal>
```